A Fully Centered First-Root Carrier
Fifteen Integers, Degree 48, 108 Gates

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1 A fully centered fifteen-integer carrier

The new endpoint uses fifteen unrestricted integer variables, has exact total degree 48, and admits a 108-gate division-free evaluator (61 multiplications, 25 additions, 22 subtractions). The retained predecessor used degree 56 and 110 gates at the same variable count. This is an ordinary arithmetic equivalence to that predecessor; the inherited native/compiler/refutation theorem, not a new consistency proof or formalized object-language derivation, supplies the connection to Con(ZFC).

1.1 Complete polynomial

The variables are

$$j, C, z, h, r, v, w, T, c, n, f, y, p_1, p_2, p_3 \in \mathbb{Z}.$$

The six numerals $\beta, K_0, M_C, M_F, A_c, B_c$ are the authenticated fixed payloads in the accompanying coefficient directory; M_F is already shifted. All names below are polynomial abbreviations, not extra witnesses:

$$\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \\
X = qW, \quad Y = qv, \quad F = (K_0 + W)C - zq_m, \\
R = (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
k = R + 1 + hE, \quad \eta = c - Yk, \quad L = EYk, \\
I = (2r + 1)c^2, \quad Q = \Delta(f^2 - 1), \quad V = f^2(cT - R) - c, \\
M = 2q^2, \quad U = MH, \quad K_{\mathrm{lo}} = 2X^2 - 3Xc - 2, \\
Z = Un + MK_{\mathrm{lo}} + H(v - q^2), \\
D_{\mathrm{err}} = (A_cF - q)^2 + (B_cC - q)^2, \\
B_{\mathrm{udget}} = k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2, \\
P_{\mathrm{os}} = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).
\end{gathered}$$

Define

$$\begin{aligned}
C_{\mathrm{new}} &= (Z + Uk)^2 + 4U(LZ - U), \\
B &= f^2 - \Delta I^2 - 1, \\
C_{\mathrm{aux}} &= V^2 - (Q - 1)y(2V + y) - 1, \\
G &= B_{\mathrm{udget}} - P_{\mathrm{os}}.
\end{aligned}$$

The one equation is

$$\boxed{P = C_{\text{new}}^2 + B^2 + C_{\text{aux}}^2 + G^2 = 0.}$$

In particular, no inequality, rational-domain variable, division, or free coefficient port occurs in the literal final polynomial.

1.2 Monicity, parity, and exact inverse maps

The pinned $\beta = 2^{1064921} - 1$ is odd and 1 (mod 3). Hence q is nonzero, even, and 2 (mod 3). Also H is nonzero and odd, and $H \equiv 3 \pmod{q}$. Thus $\gcd(H, 2q^2) = 1$ and $U \neq 0$ on every signed assignment. At a zero of P, put $t = Z/U \in \mathbb{Q}$. The new carrier is equivalently

$$t^2 + 2(2L + k)t + k^2 - 4 = 0.$$

Monicity gives $t \in \mathbb{Z}$ before any Pell or sign argument. Reducing $U \mid Z$ modulo the coprime factors M, H yields

$$v_0 = \frac{v - q^2}{M} \in \mathbb{Z}, \qquad g = \frac{K_{\text{lo}}}{H} \in \mathbb{Z}, \qquad t = n + g + v_0.$$

Reduction of the monic equation modulo 2 shows $t + k$ is even. Therefore $s = (t + k)/2$ is an integer. Equivalently, $(t + 2L + k)^2 = 4[1 + L(L + k)]$, and its integer square root is even.

The predecessor differs only in its carrier and its n coordinate:

$$C_{\text{old}} = Z_{\text{old}}^2 + (2Z_{\text{old}} - Uk)UL - U^2.$$

The universal polynomial identity

$$C_{\text{new}}(2Z_{\text{old}} - Uk) = 4C_{\text{old}}(Z_{\text{old}})$$

proves both directions. From a new zero keep every other input and set

$$n_{\text{old}} = \frac{n_{\text{new}} - g - v_0 + k}{2} = s - g - v_0 \in \mathbb{Z}.$$

Then $Z_{\text{old}}/U = s$ and $C_{\text{old}} = 0$. Conversely the old carrier is monic in Z_{old}/U; it therefore licenses the same CRT offsets at every old zero. Set

$$n_{\text{new}} = 2n_{\text{old}} + g + v_0 - k.$$

Then $Z_{\text{new}} = 2Z_{\text{old}} - Uk$ and $C_{\text{new}} = 0$. The other three residuals do not involve n and remain zero in both directions. These maps are mutually inverse on the zero sets. Divisions in the proof maps are licensed at zeros; none is hidden in the polynomial.

1.3 Exact degree and arithmetic accounting

The relevant degrees are

$$\deg U = 6, \quad \deg Z = 7, \quad \deg L = 11, \quad \deg k = 5.$$

The unique degree-24 term of C_{new} is $4ULZ$. The four residual degrees are $24, 14, 18, 16$, so $\deg P \leq 48$. Writing leading homogeneous parts with subscript 0 gives

$$(C_{\text{new}})_{24} = 4U_0^2 L_0 n, \qquad P_{48} = 16U_0^4 L_0^2 n^2,$$

where

$$q_0 = 6\beta j, \qquad X_0 = 12q_0 w, \qquad Y_0 = q_0 v,$$
$$U_0 = 8q_0^2 X_0 Y_0, \qquad L_0 = X_0 Y_0^2 (R_0 + hX_0 Y_0).$$

Because R is independent of h, the actual coefficient is

$$[j^{26} w^8 v^{10} h^2 n^2] P = 2^{58} 3^{34} \beta^{26} > 0.$$

Its exponents sum to 48; the other squared residuals have smaller degree. This proves exact degree after the huge constants are fixed.

The evaluator has 108 live binary gates and all fifteen inputs are live. Squares and scalar multiplications are charged. Exact register-cut identities, source/payload hashes, leading forms, liveness, counts, and deliberately bad mutations are checked independently of the builder, under normal and optimized Python. These finite certificates do not replace the unbounded inherited compiler/refutation theorem or assert global minimality.